Magnetism in Heard Island Rocks

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Abstract. The directions of remanent magnetization in igneous rocks from Heard Island show both negative and positive polarities. Their mean, irrespective of sign, is close to the hypothetical axial geocentric dipole field but differs from it by an amount which is about equal to the error in the determination. This divergence may not be real and requires further study, but if it is real it could be explained by a southerly movement of Heard Island during the Cenozoic as suggested by J. T. Wilson's hypothesis of earth behavior. More detailed sampling at Heard and also in the Kerguelen archipelago could provide a critical test for this hypothesis. The dispersion of directions is measured by a circular standard deviation of 11°, which is much less than the secular variation in these southern latitudes today.

1. *Introduction.* During the 1963 Heard Island Expedition, P. J. Stephenson collected 24 oriented samples and sent them to the Australian National University for magnetic study. The expedition was made under difficult and at times hazardous conditions, which limited the number of samples it was possible to collect. It is unlikely that further collections will be made from this remote and inhospitable island in the near future; and we therefore believe that despite the small number of samples it is worth while to record the results, since they may relate to the interpretation of oceanic magnetic anomalies and also suggest a possible test for J. T. Wilson's recent ideas about the drift of oceanic islands.

2. *Geology and sampling.* Geological descriptions have been given by *Philippi* [1908], *Reinisch* [1908], *Aubert de la Rue* [1929], *Mawson* [1932], *Tyrrell* [1937], and *Lambeth* [1952]. The geological sequence followed here (Table 1) closely follows Lambeth's, with the reinterpretation of that author's Drygalski agglomerates as glacial sediments [*Stephenson,* 1964]. The oldest rocks exposed are foraminiferal limestones of early Tertiary age, with intermediate hypabyssal intrusions. The limestones and intrusions are overlain unconformably by a sequence of glacial sediments with lavas and associated dikes and sills whose age is presumed to be Pleistocene. Above these lie (in some places with unconformity) the Big Ben lavas of Pleistocene to Recent age, which also have associated minor intrusions. Finally there are lavas and dikes of Recent age, which are restricted to the highest point on the island and to certain coastal areas.

Sample orientations were made by visual sighting to topographic features and the azimuths then read from maps. The sampling area is within the triangulation survey made by R. G. Dovers in 1948. Orientations are accurate to 3°. The determination of intensities of magnetization and susceptibility are accurate to 5%. The directions are accurate to better than 2°, so that the combined field and laboratory errors are 3 to 4°. Two or three samples were obtained from each of the 10 sites. Cylindrical specimens (21 mm × 21 mm, and 48 all told) were cut from the samples. The stratigraphic relations of the sampling sites are given in Table 1 and their areal distribution in Figure 1. Locality descriptions are given below.

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TABLE 1. Schematic Geological Section and Stratigraphic Position of Sampling Sites

The sites whose relative ages are known from direct geological evidence are separated from those sites with indefinite age relationships.

	Site	
	Definite Relationships	Indefinite Relationships
Coastal volcanics and associated minor intrusions (Recent)	1 2	7
Big Ben lavas and associated minor intrusions (Pleistocene to Recent)		8
	3	
Glacial sediments, lavas, and associated intrusions (Pleistocene?)	4 5	9 10
Intrusions	6	
Limestones (Lower Tertiary)		

Site 1. 53°01′S, 73°24′30″E. Limburgite dike associated with Recent cone on Rogers Peninsula. Samples collected from the gully to the south of Corinth Head.

Site 2. 53°01′S, 73°29′E. Recent lava, Saddle Point. Samples of porphyritic limburgitic basalt collected from sea cliffs.

Site 3. 53°02′S, 73°23′E. Porphyritic olivine basalt. Samples collected near the north termination of low ridge northwest of Mt. Drygalski.

Site 4. 53°01′S, 73°21′30″E. Lavas above glacial sediments. Samples of porphyritic basalt (clinopyroxene phenocrysts) collected from the crest of razorback ridge, approximately 300 ft above sea level.

Site 5. 53°01′S, 73°21′30″E. Lava above glacial sediments. Samples of porphyritic basalt (clinopyroxene phenocrysts) collected from lava flow directly overlying tillite, approximately 150 ft above sea level.

Site 6. 53°02′S, 73°27′E. Porphyritic micromonzonite hypabyssal intrusion in Lower Tertiary limestones, unconformably overlain by glacial sediments. The limestones are quasi-horizontal and the intrusion has probably been little moved since emplacement. Samples were collected from the chilled marginal phase adjacent to contact with limestones which occurs beneath bluff east of the Baudissin glacier.

Site 7. 53°02′S, 73°23′E. Dike associated with the eroded volcanic cone of Mt. Drygalski (Recent ?). Samples of porphyritic limburgitic basalt were collected from the northwest slopes

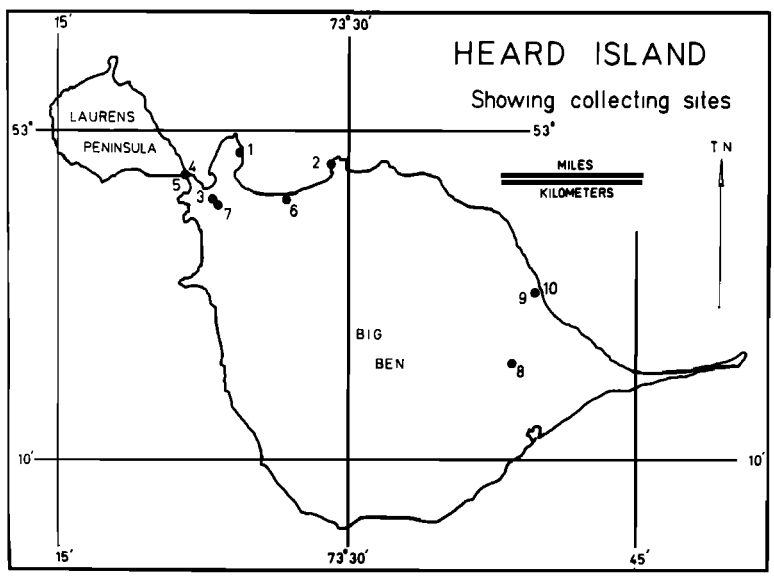


Fig. 1. Scale lengths are 5 miles and 8 km.

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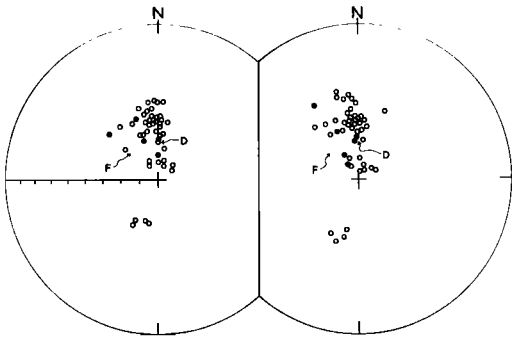


Fig. 2. Directions observed in each specimen are plotted on the upper hemisphere of equal-area projections—NRM on the left, and after magnetic cleaning (150 oe (peak) alternating field) on the right. Circles are north-seeking directions and dots south-seeking. The heads of the arrows *F* and *D* are the directions (upward) of the present and axial geocentric dipole fields. The lower four directions are from site 7.

of this mountain, approximately 300 ft below its summit.

Site 8. 53°07′S, 73°38′30″E. Dike intruding pillow lavas and glacial sediments. Samples of porphyritic limburgitic basalt collected near the crest of nunatak ridge.

Site 9. 53°05′S, 73°40′E. Sill (?) in pillow lavas and glacial sediments. Samples of porphyritic olivine basalt collected from cliffs behind the coast between the Round Hill and Scarlet Hill glaciers.

Site 10. 53°05′S, 73°40′E. Sill (?) in pillow lavas and glacial sediments. Samples of porphyritic olivine basalt collected near site 9, but lower down, farther east, and in a small gully.

3. *Directions of remanent magnetization.* The directions of natural remanent magnetization (NRM) of specimens at eight sites have directions with polarity like that of the present field (normal or negative polarity); those at site 6 have positive polarity (Figure 2). At site 7 the directions are divergent from the axis through the negative and positive directions.

The stability of the remanence has been studied by alternating field demagnetization methods [*Thellier and Rimbert,* 1955] using the apparatus and procedures described by *Irving et al.* [1961]. An intensity decay curve from a negatively polarized specimen (site 9) shows a smooth decrease, and the direction is little changed (Figure 3). A curve from a positively polarized specimen (site 6) falls slowly at first, then rapidly, and direction changes of a statistical nature occur. The difference in the curves in the range up to 300 oe is attributed to the presence of viscous components imposed by the present field, and in order to demagnetize these, all specimens have been magnetically cleaned in 150 oe (peak). The distribution of directions after cleaning does not differ appreciably from that of the NRM directions (Figure 2). These observations suggest that the NRM consists predominantly of a single stable component. Small soft components may be present but these are demagnetized by cleaning. In view of the igneous origin of the rocks and their fresh condition, it is reasonable to suppose that the stable magnetization is thermoremanent (TRM) in origin and was acquired at cooling. The absence of any anisotropy effects (see section 6) leads us to believe that the initial alignments were accurately parallel to the ambient field. The mean site directions, obtained by first averaging the specimen direction from each sample and then averaging the sample means, are listed in Table 2.

The directions in the dike at site 7 are puzzling. The discordance is not likely to be due to recently imposed viscous components since magnetic cleaning does not alter the directions substantially (Figure 2). It is unlikely that the directions are part of a transition between reversals since the dike is of probable Recent age.

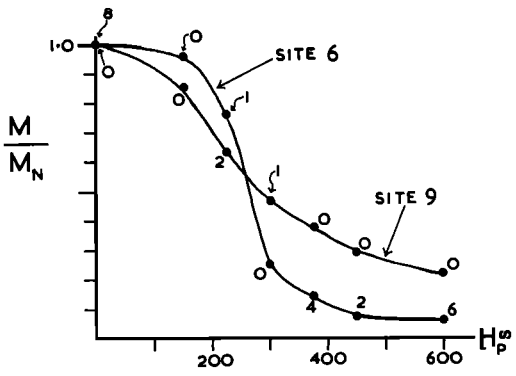


Fig. 3. Alternating field demagnetization curves. The ratio of the intensity *M* after treatment in a field of strength $H_p^{\sim}$ to the natural remanent intensity is plotted as a function of $H_p^{\sim}$. The numbers at each experimental point are the deviations (in degrees) in excess of the measurement errors (2°) from the direction after treatment in 150 oe, the cleaning field used.

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TABLE 2. Summary of Results

	N	S	R	D,	I	Site Pole	M_N and χ, cgs/cm^3 × 10^4
Site 1	2	4	1.99(8)	358,	−59	N77, 066E	160.5, *4.2*; 37.3, *3.6*
Site 2	2	4	1.99(9)	358,	−57	75, 068	28.4, *1.6*; 202.0, *2.5*
Site 3	3	6	2.97	008,	−77	76, 239	63.9, *2.8*; 38.4, *1.5*; 65.9, *1.7*
Site 4	2	4	1.97	353,	−61	78, 048	49.1, *39.0*; 68.7, *42.6*
Site 5	2	4	1.99(9)	354,	−44	62, 062	56.4, *39.0*; 54.8, *38.1*
Site 6	3	6	2.99	161,	+65	76, 002	49.9, *31.2*; 11.0, *36.4*; 13.2, *21.5*
Site 7	2	4	1.96	183,	−59		51.0, *10.8*; 58.7, *11.0*
Site 8	2	4	1.99(8)	014,	−83	65, 246	93.3, *1.7*; 37.5, *13.8*
Site 9	4	8	3.98	354,	−56	73, 057	25.4, *1.4*; 33.6, *2.6*; 29.0, *2.1*; 28.9, *2.5*
Site 10	2	4	1.99(6)	342,	−56	69, 028	42.2, *3.6*; 41.7, *5.2*

N is the number of oriented samples and S the number of cylinders measured. R is the magnitude of the site resultant (sample unit weight, after magnetic cleaning in 150 oe H_p~, third place given in parentheses when impossible situation $R = N$ arises from rounding off) and D, I its mean direction. The northern hemisphere site poles are given. In the last column the intensity of NRM M_N and the susceptibility χ (italics) for each sample (N) is given.

The discordance, however, is that expected from an 180° orientation error in azimuth. Because there is good agreement between two independently oriented samples, such an error is not likely, but since the bearing taken for both samples was to the same mountain peak the possibility of such a mistake cannot be discounted. It is not possible to resample in the foreseeable future, and because of this uncertainty the site 7 directions are not further considered.

The directions after magnetic cleaning (150 oe H_p~) at the remaining nine sites have been combined, irrespective of sign, using *Fisher*'s [1953] method and the two-tier statistics of *Watson and Irving* [1957]. The results from the positive site have been treated as negative by reversing the directions through 180°.

Over-all mean direction D_m, I_m = 353, −62
Resultant (sites [9] unit weight) R_b = 8.808
Resultant (samples [22] unit weight) R_s = 21.449
Fisher's estimate of precision (sites unit weight) k = 42
Over-all precision k' = 400
Within-site precision k_w = 113
Between-site precision k_b = 53
Radius of the circle of confidence at P = 0.05 (sites unit weight) a_{95} = 8°
Radius of the circle of confidence at P = 0.05 (two-tier analysis) a'_{95} = 7°

4. *Discussion of direction results.* The time covered by the samples is not accurately known, but if the tentative age assignments of Table 1 are reasonable, then sites 1 to 5 and 7 to 10 cover a period extending back from the present to, say, 10^5 years or even as much as 0.5 m.y. Site 6 is post-Paleocene; it may be Tertiary or early Pleistocene, and it further extends the time coverage. The mean direction (Figure 4) differs from the present field. The departure from the geocentric axial dipole field (8°) is equal to the error obtained by giving sites unit weight; if the error given by the two-tier analysis is used, the departure is significant.

A further procedure is to calculate paleomagnetic poles and to compare these with the present geographical pole. In these comparisons use is made of statistical models (designated A and B) of the earth's fields. When model A is used (in model A the main geocentric dipole field is imagined to be perturbed by randomly directed fields at the earth's surface [*Irving and Ward*, 1964]), a pole is calculated from the over-all mean direction [*Creer et al.*, 1957]. When model B is used (in model B it is assumed that the surface field fluctuations result from oscillations of the geocentric dipole itself [*Creer et al.*, 1959]), poles are calculated from the mean site directions and then are combined by Fisher's method to give a mean pole. These are the results.

Model A: pole 79°N, 45°E; dp = 10°, dm = 12°
Model B: pole 81°N, 41°E; K = 22, A_{95} = 11°

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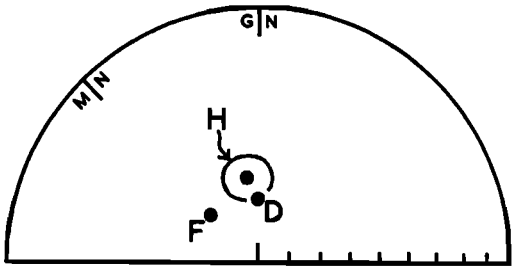


Fig. 4. Mean direction of Heard Island samples and circle of confidence (H) compared with the present (D) and axial geocentric dipole field (D). Plotting an upper hemisphere of an equal-area projection.

dp and dm are the errors ($P = 0.05$) in the direction of and perpendicular to the paleomeridian, K is the precision of site poles and A_{95} the error ($P = 0.05$). Model A pole departs marginally from the geographic pole; model B pole does not.

The results are potentially applicable to the hypothesis of *Wilson* [1963], which relates to the origin of the transverse oceanic ridges. The mid-oceanic ridges often have associated transverse ridges running toward the continental margins. Wilson suggests that the transverse ridges mark the path of continental movements. His evidence for this is that the ages of the oldest rocks on the islands of the transverse ridges increase with distance from the mid-oceanic ridges, and this fact, he contends, provides an estimate of the rates of movement. For instance, on this hypothesis Heard Island has moved 17° southward since the early Tertiary at an average rate of 3 cm/yr. Since the motions supposedly recorded by the islands of the transverse ridges have largely occurred in the Cenozoic, such motions, to be detected paleomagnetically, must have a strong N-S component. The most promising islands for study are the Kerguelen archipelago and Heard. (The supposed longitudinal motions of the islands of the transverse ridges of the Atlantic would not be seen.)

It is not possible on the basis of present sampling to carry out a rigorous test of Wilson's suggestion, but it is noteworthy that the departure of the best estimate of the mean (a blurred average of the field for the period in question) is in the correct sense for his hypothesis (Figure 4)—the paleolatitude given by the best estimate of the mean inclination is 43°, placing Heard about 10° farther north than at present. Clearly no case for such a motion can be made from the present evidence, yet it seems possible that a detailed paleomagnetic survey of Heard and Kerguelen could provide a useful test which would be particularly effective if it were made side by side with a radiometric age determination of the age of the lavas.

The between-site dispersion (freed of that within sites) may be taken as an estimate of the dispersion of the field over the period spanned by the sampling sites. This is given by a circular standard deviation of 11°, which is less than the dispersion of the 1945 field [*Creer*, 1962] in these latitudes (s.d. 15°) but similar to the mean for both hemispheres (s.d. 11°).

5. *Remanent intensities and susceptibilities (units of 10^{-4} cgs/cm^3).* The intensities of natural remanence form a single group with an arithmetic mean of 55.9. The distribution is skewed but becomes symmetrical when logs are taken. The log mean is 45.4 (standard deviation $\log_{10} = 4.284$). This value is comparable to those obtained by *Cox and Doell* [1962] for the basalt taken from hole EM7 in the ocean flow of the east Pacific (arithmetic mean 53.8, log mean 50.5), by *Doell and Cox* [1963] for the historic lavas of Hawaii (arithmetic mean 73.2, log mean 70.2×10^{-4}), and by *Creer* [1964] for the basalts of Tristan and Inaccessible islands (arithmetic mean 82.8, log mean 74.6). It may be compared also with the result of *Serson and Hannaford* [1957], who concluded from their analysis of aeromagnetic profiles over oceans that 'under the deep oceans, there are many bodies magnetized to an intensity of at least 0.005 cgs.'

Susceptibilities were measured in a steady field of 0.54 oe. They fall into two groups with log means of 2.4 and 25.1, respectively. In over 90% of the samples the remanence component exceeds that induced by the present field by a factor of 2 or more, and in over 60% of the samples by a factor exceeding 10. If these samples are approximately representative of oceanic igneous material in the vicinity of Heard, the effect of remanence on geomagnetic anomalies observed nearby will greatly exceed that due to induction by the present field. In order that detailed comparisons may eventually be made, values for each sample are recorded in Table 2.

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6. *Thermoremanent magnetization.* The following experiments were made in an attempt to estimate the strength of the field in which the samples originally cooled and to study the possibility that, owing to anisotropy, the reported directions do not correspond accurately to the direction of that field. Eleven specimens (representatives from all sites) were heated to 600°C in a nitrogen atmosphere and cooled in a field of 0.59 oe to give a TRM of magnitude M_T. The range, arithmetic mean, and logarithmic mean of these samples, together with the NRM intensities of the same specimens, are listed in Table 3. The strength H_P of the ancient field may be estimated from the relationship $H_P = M_N H/M_T$, where H is the strength of the field in which M_T is acquired [*Thellier,* 1938]. We get 0.31 oe from the arithmetic mean and 0.29 oe from the logarithmic mean for the field intensity at the time the samples cooled. Three assumptions are made: first, that M_N varies linearly with strength field in the low field ranges in question; second, that there has been no viscous decay of the natural remanence (some viscous decay is almost bound to have occurred so that H_P is a lower estimate); third, that in the acquisition of TRM in the laboratory no physico-chemical changes occurred to enhance or diminish remanence. A refinement used by *van Zijl et al.* [1962] decreases the effect of viscous decay. In this the remanences compared are those after demagnetization by alternating magnetic fields of the low-stability components —those most likely to have decayed. The demagnetizing field used is 150 oe. The estimates of the mean field then become 0.43 and 0.37 oe for the arithmetic means and logarithmic mean, respectively. Although crude, these estimates indicate that the mean strength of the field (0.4 oe) during the period concerned was comparable to that at present (0.49 oe).

TABLE 3. Comparison of Intensities

	Range	Arithmetic Mean	Log Mean
TRM	40–256	138	114
TRM′ (after partial demagnetization)	24–221	87	71
NRM	25–201	74	59
NRM′ (after partial demagnetization)	21–167	55	44
NRM/TRM	0.31–1.08	0.53	0.49
NRM′/TRM′	0.19–1.46	0.73	0.63

Intensities given $\times 10^4$ cgs/cm³. Demagnetized in alternating magnetic field of 150 oe (peak).

In these experiments the specimens were placed in the furnace in the same orientation with the magnetic field that they had before they were extracted from the outcrop. The TRM's were in all cases directed along the applied field, whose direction was (taking magnetic north as zero azimuth) 000, −65.5. The mean direction of the 11 specimens is 001, —66, the precision $k = 3518$, the circular standard deviation is 1.4°, and the radius of the circle of confidence $a_{95} = 0.7°$. The mean coincides with the direction of the applied field. The scatter is accounted for entirely by measurement errors (see section 2) so that the effect of anisotropy on the remanence directions is negligible.

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References

Aubert de la Rue, E., Contribution à l'étude géologique de l'Ile Heard, *Compt. Rend., 189,* 129–131, 1929.

Cox, A., and R. R. Doell, Magnetic properties of the basalt in hole EM7, Mohole project, *J. Geophys. Res., 67,* 3997–4004, 1962.

Creer, K. M., The dispersion of the geomagnetic field due to secular variation and its determination for remote time from paleomagnetic data, *J. Geophys. Res., 67,* 3461–3476, 1962.

Creer, K. M., Palaeomagnetic measurements on lavas from Tristan and Inaccessible Island, in *Volcanological Report on Tristan da Cunha, Phil. Trans. Roy. Soc. London, A, 256,* 569–573, 1964.

Creer, K. M., E. Irving, and A. E. M. Nairn, Palaeomagnetism of the Great Whin Sill, *Geophys. J., 2,* 306–323, 1959.

Creer, K. M., E. Irving, and S. K. Runcorn, Geophysical interpretation of palaeomagnetic directions from Great Britain, *Phil. Trans. Roy. Soc. London, A, 250,* 144–156, 1957.

Doell, R. R., and A. Cox, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows, *J. Geophys. Res., 68,* 1997–2009, 1963.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 295–305, 1953.

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Irving, E., P. M. Stott, and M. A. Ward, Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag., 6,* 225–241, 1961.

Irving, E., and M. A. Ward, A statistical model of the geomagnetic field, *Pure Appl. Geophys. (Milan), 57,* 25–30, 1964.

Lambeth, A. J., A geological account of Heard Island, *J. Proc. Roy. Soc. N. S. Wales, 85,* 14–24, 1952.

Mawson, D., The B.A.N.Z. Antarctic Research Expedition 1929–31, *Geogr. J., 80,* 105–106, 1932.

Philippi, E., Geologie de Heard-Insel, Deutsche Südpol. Exped. 1901–1903, *Geogr. u. Geol., 2,* 241–250, 1908.

Reinisch, R., Gesteine der Heard Insel, *Geogr. u. Geol., 2,* 251–263, 1908.

Serson, P. H., and W. L. W. Hannaford, A statistical analysis of magnetic profiles, *J. Geophys. Res., 62,* 1–18, 1957.

Stephenson, P. J., Some geological observations on Heard Island, *Proc. Scientific Committee for Antarctic Research, Symposium on Antarctic Geology,* Capetown 1963, 1964 (in press).

Thellier, E., Thesis, University of Paris, 1938.

Thellier, E., and Francine Rimbert, Sur l'utilisation, en paléomagnétisme, de la désaimantation par champs alternatifs, *Compt. Rend., 240,* 1404–1406, 1955.

Tyrrell, G. W., The petrology of Heard Island, *B.A.N.Z. Antarctic Res. Exped. 1929–1931, Reports, Ser. A., 2,* 29–56, 1937.

van Zijl, J. S. V., K. W. T. Graham, and A. L. Hales, The palaeomagnetism of the Stormberg lavas, 2, *Geophys. J., 7,* 169–182, 1962.

Watson, G. S., and E. Irving, Statistical methods in rock magnetism, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl., 7,* 289–300, 1957.

Wilson, J. T., Hypothesis of earth's behaviour, *Nature, 198,* 925–929, 1963.

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